

Final Report (CSE299.4)

Team Information

ID	Name	Role/Contribution
2221080642	Tashfiq Mahmud Niloy	Mostly done all the Backend Part. That is with programming.
2211951042	Tamanna Akter Mou	Helped Tashfiq with programming and merge weekly report
2221838642	Ihtishamul Haq Shimanto	Management and QA
2211659042	Tirana Mahbub Meem	Inserted UI/UX and Audio

Project Identification

Detail	Description
Project Title	Vapor Drift
Type	Endless Runner Game
Platform	PC
Game Engine	Unity
Programming Language	C#
Group Number	05
Date Submitted	15th December 2025

Executive Summary

"Vapor Drift" is a successfully completed, fast-paced, stylized 3D endless runner game developed using the Unity engine and C#. The project successfully implemented all core mechanics, including three distinct difficulty modes and a comprehensive UI with adjustable sensitivity, designed for the PC platform. The player's objective is to collect 100 coins while navigating a low-poly urban environment. The game employs a clean, premium business model, utilizing an upfront download fee and non-intrusive cosmetic In-App Purchases (IAPs) for revenue, strictly avoiding advertisements.

Problem Statement and Solution

1 Problem

The existing market for endless runner games is often saturated with titles that rely on disruptive and frequent in-game advertisements, severely degrading the player experience and prioritizing monetization over gameplay quality.

2 Solution (Vapor Drift)

Vapor Drift offers a superior, skill-focused arcade alternative. The project provides an uninterrupted, premium experience by implementing a "No Ads" business model. The game is further distinguished by its low-poly 3D urban aesthetic and a focus on core player skill and mastery.

Technical Design and Implemented Features

1. Game Engine and Tools

- Engine: Unity
- Language: C#
- Art Style: Low-Poly 3D, featuring bright, simplified urban environments and character models.

2. Core Game Mechanics and Rules

The fundamental gameplay loop and rules have been fully implemented and tested:

- Win Condition: Collecting exactly or more than 100 coins.
- Loss Condition (Collision): Game ends after the player sustains the 5th collision with an obstacle (cars).
- Loss Condition (Time-out): Game ends if the clock reaches zero in the time-based levels.

Player control is fully functional using PC keyboard inputs: WASD controls translational movement; Z/X enables precise turning/drifts, and SPACE executes a jump.

3. Implemented Features

Feature	Implementation Detail
Difficulty Modes	Three selectable modes: No Timer (pure endurance), Easy Level (2 minutes to collect 100 coins), and Hard Level (30 seconds to collect 100 coins).
Configurable Settings	Audio and Sound sliders are functional in the Main Menu. A high-precision Sensitivity slider (range up to 10.0) is integrated into the Pause Screen.
Heads-Up Display (HUD)	The HUD accurately displays the Timer, Coin Counter, and the Health Indicator (bandaged face icon) for tracking the 5-hit limit.
Game State Control	Full functionality for Main Menu navigation and in-game Pause/Resume/Restart mechanics is complete.
Rules Display	A dedicated "RULES" screen is fully implemented to clearly state all objectives, loss of conditions, and controls.

Project Management and Division of Labor

The project successfully concluded by adhering to defined roles and ensuring comprehensive quality assurance:

- **Programming (Tashfiq & Tamanna):** Delivered the core movement and physics, procedural track generation, coin collection system, and implemented the precise logic for the three difficulty modes in C#.
- **UI/UX and Audio (Tirana):** Created and integrated all user interfaces (Menu, Pause, Rules), implemented audio controls, and ensured the sensitivity slider correctly adjusted player input variables within the Unity environment.
- **Management and QA (Ihtishamul):** Formalized the official project ruleset, managed task distribution, and led comprehensive testing to validate all win/loss conditions (100 coins, 5 bumps, time limits).

Business and Monetization Strategy

The game adopts a non-disruptive, premium approach to revenue generation:

1. **Primary Revenue: Initial Download/Purchase Fee.** This upfront cost is justified by the commitment to an ad-free player experience.
2. **Secondary Revenue (IAPs):** Limited strictly to cosmetic skins. This ensures that all purchases are purely aesthetic, reinforcing the game's core value as a skill-based, non-pay-to-win title.

Conclusion

"Vapor Drift" is a complete project that successfully integrates a distinctive low-poly aesthetic with a solid, skill-based endless runner framework developed in Unity and C#. All defined features, including the variable difficulty levels and customizable controls, have been successfully implemented. The game is ready for submission and distribution, offering a quality product supported by a sustainable, player-friendly business model.